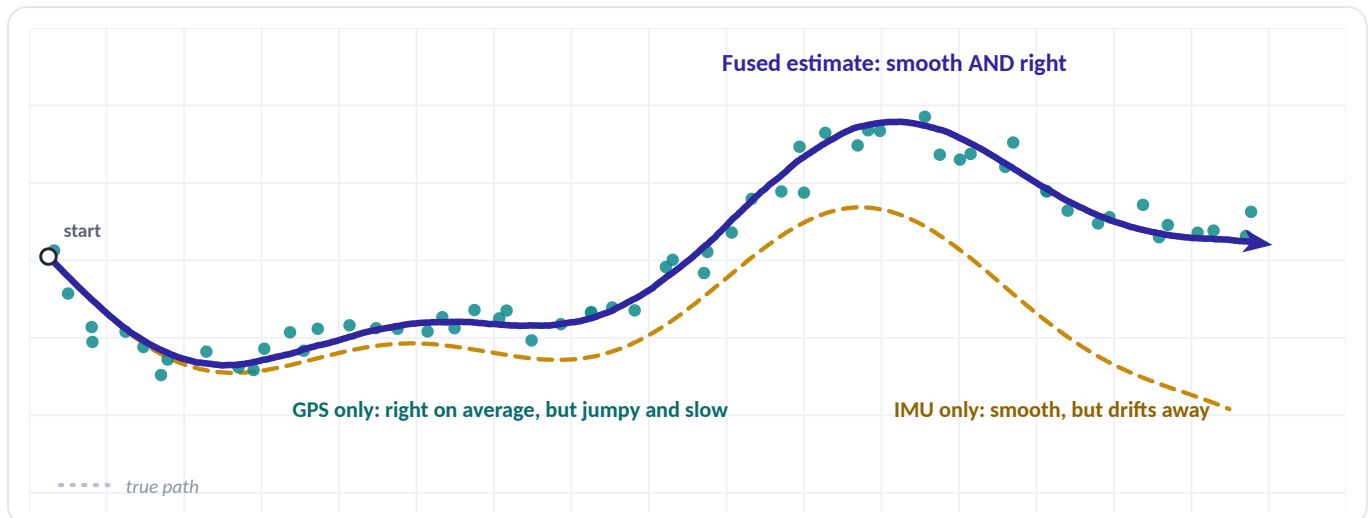




Where are we, which way are we pointing, and how fast are we going?

This team answers that question hundreds of times every second. Everything else on the aircraft depends on the answer.



Every sensor lies a little, in its own way. Combine them properly and the truth comes out. That combining is our job.

WHAT WE DO

- Write the **real-time logic** that turns raw sensor numbers into the aircraft's **state**: attitude, position, speed, height
- Decide **where to fly next**: waypoints, route, targets
- Hand both to **flight control** every loop. Bad estimate, bad flight
- Produce the **measured numbers** NFC scores us on
- Design a **physical air-data board**: our own airspeed channel

WHY IT MATTERS

NFC scores how closely we fly to our own predictions. The **tightest cut-offs (10%)** are on **airspeed and altitude**. Those numbers come from us.

The Kalman filter flew to the Moon on **Apollo**. Today it runs in your phone, every drone and every self-driving car. You will learn how.

WHAT WE HAND FLIGHT CONTROL, 100+ TIMES A SECOND

ROLL 12.4 degrees	PITCH 3.1 degrees	HEADING 274 degrees	AIRSPEED 18.6 m/s	ALTITUDE 62.0 m	CLIMB +1.2 m/s
--------------------------------	--------------------------------	----------------------------------	--------------------------------	------------------------------	-----------------------------

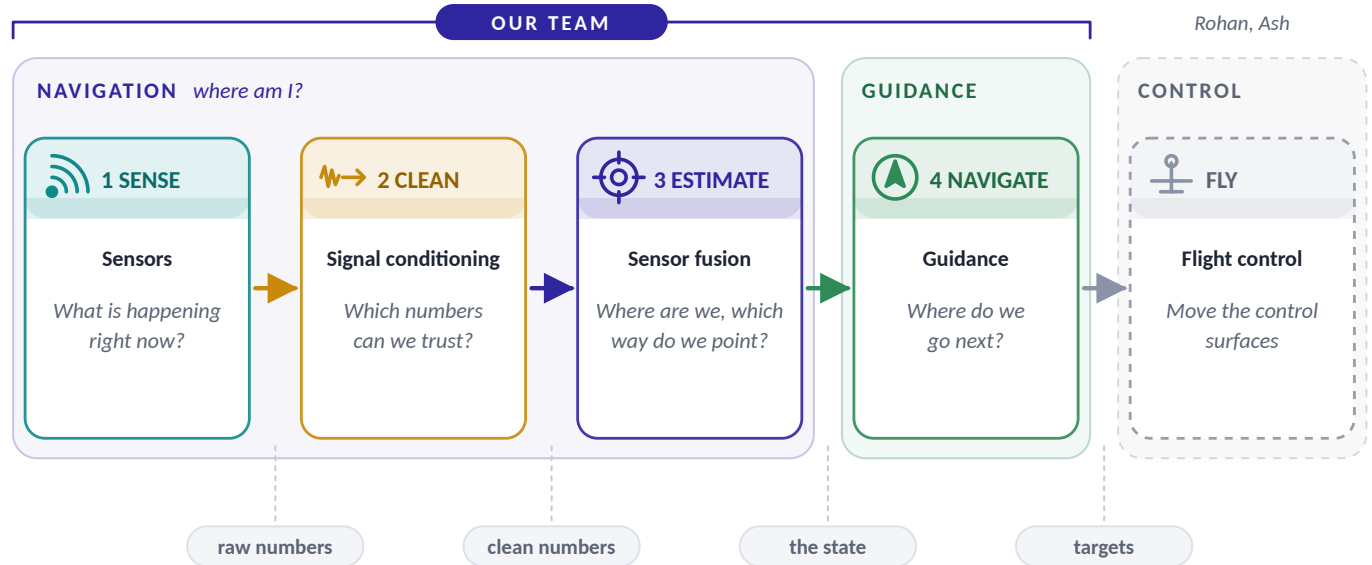
Plus position and velocity. Get any one of these wrong and the aircraft flies wrong.

IN THIS DOCUMENT

[p2 The big picture + tools](#)[p3 How it works, in detail](#)[p4 How we could build each block](#)[p5 Your to-do list](#)[p6 Pick a track + our plan](#)

1 The big picture

Industry calls this **GNC**: Guidance, Navigation and Control. We are the **G** and the **N**.



Rohan, Ash

- **Navigation** = where am I? Blocks 1 to 3. Sensors in, one best guess out.
- **Guidance** = where should I go? Block 4. Turns the mission into targets.
- **Control** = how do I get there? Flight control team. They trust our numbers completely.
- The colours stay the same for the rest of this document, including your to-do list.

DOWNLOAD NOW

- 1 MATLAB + Simulink**, free through the uni
Add: Signal Processing, Sensor Fusion and Tracking toolboxes
studentsoftware.unimelb.edu.au · new to it? [MATLAB Onramp](#)
 - 2 LTspice**, circuit simulator (free)
Simulate a filter before you build it
analog.com/ltspice
 - 3 KiCad**, schematic + PCB design (free)
Draw the air-data board
kicad.org/download
 - 4 Git** + a GitHub account
Where all our work lives
git-scm.com/downloads
- + **Optional:** Python 3 + VS Code, if you prefer Python to MATLAB · python.org/downloads · code.visualstudio.com/download

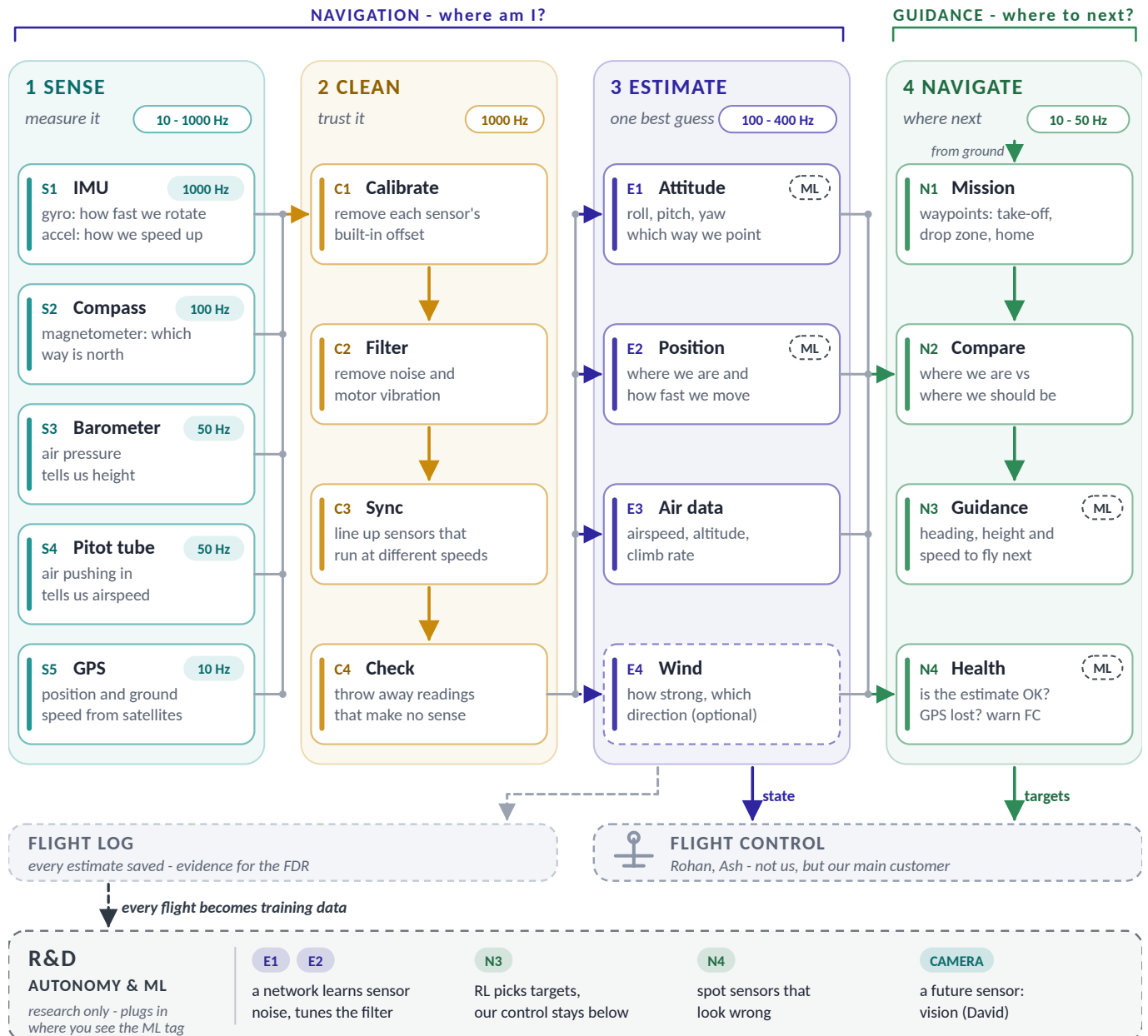
KEYWORDS TO SEARCH

- SENSE** MEMS IMU GNSS pitot-static ISA atmosphere
- CLEAN** low-pass filter anti-aliasing op-amp Sallen-Key ADC moving average IIR filter
- ESTIMATE** sensor fusion complementary filter Kalman filter EKF quaternions AHRS INS
- NAVIGATE** GNC waypoint navigation cross-track error L1 guidance NED frame
- R&D** reinforcement learning learned inertial odometry anomaly detection computer vision

Related subjects: Signals and Systems, Control Systems, Probability, Linear Algebra, Analog Electronics.

2 How it works, in detail

Same four blocks, opened up. Grey lines carry data. Every block has an ID (S1, C2, E1...). Your to-do list points back to them. Dashed = research, not flight.

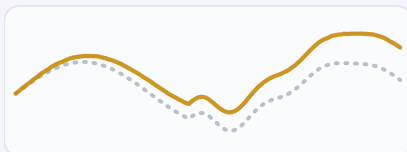


- **Fast and slow sensors.** The IMU gives 1000 readings a second, GPS only 10. Fusion fills the gaps.
- **Every stream passes C1 to C4** before we trust it.
- **Real-time:** the estimate must be ready before flight control's next loop. Late is the same as wrong.

THE CORE IDEA IN ONE PICTURE · E1 ATTITUDE

Gyro only

smooth, but drifts



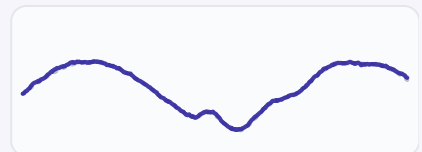
Accelerometer only

right on average, but noisy



Both, fused

smooth and right



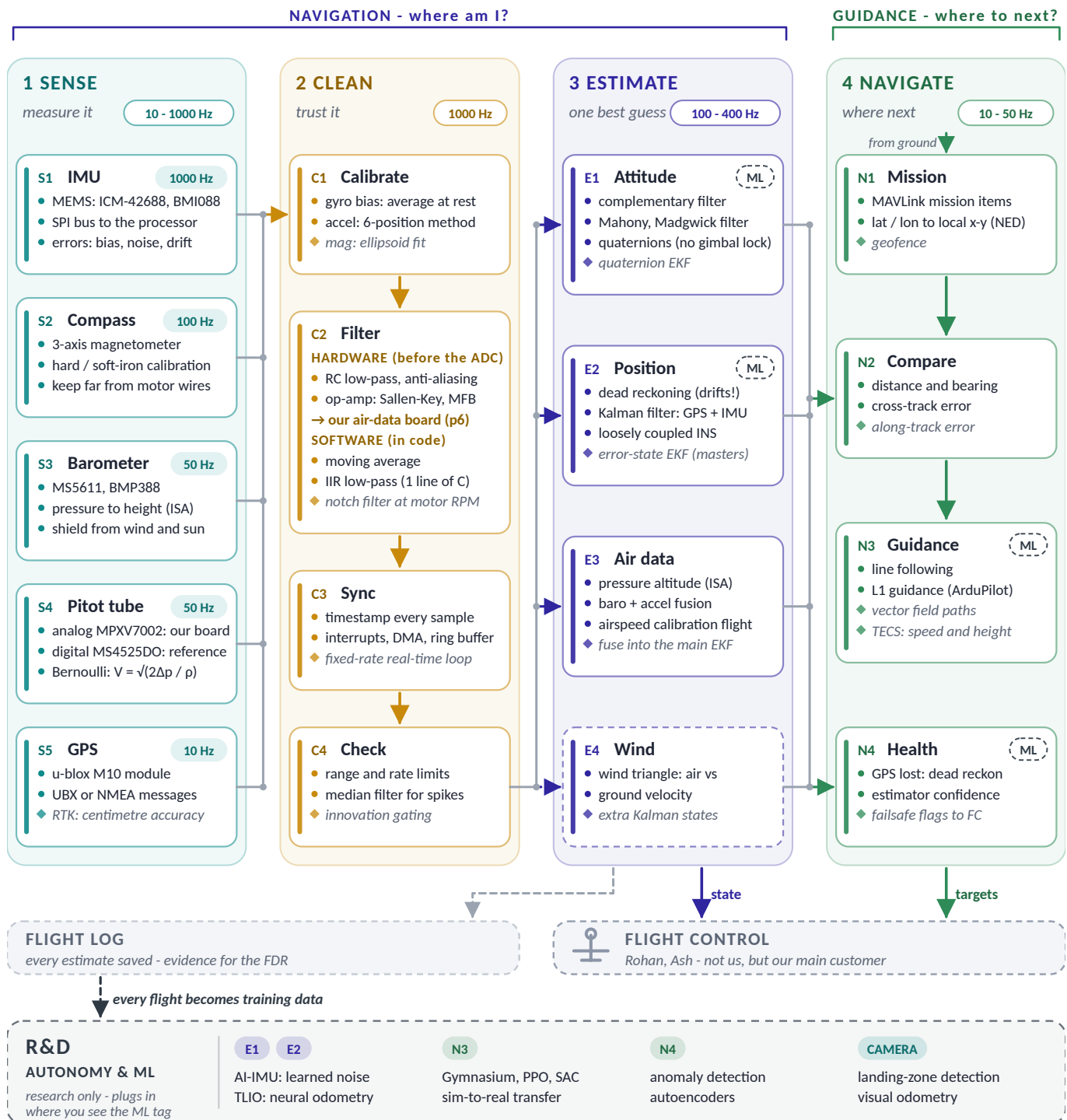
dotted line = true angle

The whole filter is one line of code: $\text{angle} = 0.98 * (\text{angle} + \text{gyro} * \text{dt}) + 0.02 * \text{accel_angle}$

3 How we could build each block

Same layout again. Each block now lists real ways to build it. Start with the simple ones.

- undergrad, start here
- advanced, masters level



C2 shows the split you will see everywhere: some filtering happens in **hardware** (resistors, capacitors, op-amps) before the signal is sampled, the rest in **software** after. Knowing which goes where is a real design decision. Our air-data board (page 6) is where this becomes real: pitot sensor, op-amp filter, then into the autopilot.

WORDS YOU WILL SEE

IMU - gyro + accelerometer on one chip
Bias - an offset a sensor always adds
AHRS - estimates attitude and heading
NED - North-East-Down, our local frame

MEMS - tiny moving parts etched in silicon
ISA - standard air model: pressure to height
INS - IMU + GPS combined for position
MAVLink - aircraft-to-ground messages

GNSS - satellite positioning (GPS is one)
Quaternion - 4 numbers that store a rotation
EKF - Kalman filter for non-linear maths
L1, TECS - ArduPilot's path and energy control



4 Your to-do list: start here

About **6½ hours**, everyone does this. No hardware needed, just watch and read. Colours and IDs match the diagrams.

YOUR 10 HOURS OVER THE BREAK

Sense 2¼ h		Estimate 2¼ h			Your track (p6) 3½ h
Sense 2¼ h	Clean 1 h	Estimate 2¼ h	Navigate ¼ h	Your block ¾ h	Your track (p6) 3½ h
SENSE S1 S2 How the sensors actually work 1 h approx Know what each chip measures and how it lies: bias, noise, drift. How MEMS accelerometers, gyros and magnetometers work (video, ~10 min) VectorNav Inertial Navigation Primer: read the sensor sections					
SENSE S3 S4 S5 Airspeed, height and GPS 1¼ h approx Airspeed and altitude are what NFC scores. GPS is our only absolute position. NASA: pitot-static tube (5 min read) Bartosz Ciechanowski: GPS, fully interactive (best visuals on the internet)					
CLEAN C2 Filtering basics 1 h approx Why we filter before sampling, and the first code anyone writes on raw data. Both tracks need this. TI Precision Labs: aliasing and anti-aliasing filters (13 min) DSP Guide ch. 15: moving average filters · ch. 19.2: single-pole (IIR) filter					
ESTIMATE E1 What sensor fusion is ¾ h approx The core idea of this team, explained with great animations. MATLAB Tech Talk: What is sensor fusion? (12 min) Fusing a mag, accel and gyro to estimate orientation (16 min)					
ESTIMATE E2 The Kalman filter 1 h approx The most important algorithm on this team. Pictures first, maths second. MATLAB: Understanding Kalman Filters, parts 1 to 3 (~20 min) Tim Babb: How a Kalman filter works, in pictures					
ESTIMATE E2 N4 How real autopilots do it ½ h approx GPS and IMU together, and the estimator we will fly first. MATLAB: Fusing a GPS and IMU to estimate pose (14 min) ArduPilot: Extended Kalman Filter overview					
NAVIGATE N3 Following a line between waypoints ¼ h approx How a fixed-wing aircraft steers itself onto the route. ArduPilot Plane: navigation tuning (L1 controller)					
YOU PICK ONE Choose your track and the block you want to own ¾ h approx Pick a track on page 6 and one block from page 3. Write 5 dot points: what it does, one way to build it, one question you still have. Bring it to the first meeting after the break.					

Do it well. Pause and sketch the diagrams yourself. Write notes in your own words. Stuck? Ask in the chat, someone else is stuck too.
 You do not need to understand every equation, just the idea behind it.
You will learn all of this in your degree at some point. May as well learn it now, on a real aircraft.

5 Pick a track

About 3½ hours. Choose one, or both if you are keen. This is where the best engineers on the team will come from.

HARDWARE S4 C2 Design the air-data board

1. **Understand the op-amp filter**
The world's most famous op-amp filter: Sallen-Key (video)
TI: Sallen-Key low-pass circuit, 6 pages
2. **Read the sensor**
NXP MPXV7002 datasheet, pages 1 to 3: pressure in, voltage out
3. **Design it**
ADI Analog Filter Wizard: pick a cutoff, get the part values
4. **Simulate it in LTspice**
LTspice for beginners (ADI, 3 videos)
How to make a Bode plot in LTspice
5. **Draw it in KiCad**
Getting Started in KiCad (official)
Phil's Lab: full KiCad PCB tutorial (watch the schematic part)

You finish with: a simulated filter and a first schematic. We order boards after the break, about \$10 for 5.

ALGORITHM E1 E2 Build the estimator

1. **Rotations and quaternions**
3Blue1Brown: quaternions, explained interactively (~30 min)
2. **Filtering in MATLAB**
Signal Processing Onramp: free, in the browser
3. **Watch a real filter being built**
Phil's Lab: complementary filter, EKF theory, EKF in code
4. **Kalman filters, hands-on**
Kalman and Bayesian Filters in Python (runs in the browser)
kalmanfilter.net: worked examples, numbers first

You finish with: a filter running on sample data, ready for our first flight logs.

WANT MORE? MASTERS LEVEL

- Intro to inertial navigation, Woodman · [Cambridge report](#)
- Measure sensor noise: Allan variance · [MathWorks](#)
- How PX4 tunes its EKF · [PX4 docs](#)
- Small Unmanned Aircraft, Beard and McLain · [ch. 8 and 10, free draft](#)
- Active low-pass filter design, TI · [SLOA049, 22 pages](#)
- Madgwick orientation filter · [paper](#), [C library](#)
- Error-state Kalman filter, Solà · [arXiv 1711.02508](#)

R&D · AUTONOMY & ML research only, see the ML tags on pages 3 and 4

Start with [Gymnasium](#) and [Spinning Up in Deep RL](#). Papers: [AI-IMU dead-reckoning](#), [TLIO](#). Full roadmap in Shangrui's R&D document.

Rule: nothing from R&D flies without bench evidence and an Engineering Manager decision.

AFTER THE BREAK, THIS LETS US

- Choose our sensors and write their requirements
- Order and test air-data board v0 against a digital airspeed sensor
- Fly ArduPilot's EKF first and log everything
- Own the air-data numbers that go into the FDR
- Build our own estimator and run it on the same logs
- Give every block on page 3 an owner

HOW WE COULD BUILD IT

A • Autopilot as-is

ArduPilot or PX4 runs its own EKF
+ flies first time, proven
- black box, little learning

B • Autopilot + ours beside it

fly theirs, run ours on the logs, compare
+ safe, and we really learn
- two systems to keep in sync

OUR PICK

C • Fully custom

our sensor board, our estimator in flight
+ full ownership
- highest risk, longest path

Start with B. Move towards C as our estimator earns trust on real flight data.